



Remote Control Robot Spider

Spiderpeople Inc.

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Headline

Content

The 30-second pitch and the longer presentation should each cover the following topics:

- 1) problem being solved or goal,
- 2) use case, i.e., what the system does from the perspective of the user, and
- 3) overview of your design.

Pitch

Pitch

- Goal: Walking controllable hexapod with multiple gaits
- Use case: Novelty. Fun way to learn about gaits and robots
- Overview: Amazon hexapod kit controlled by stm32. Communicates to a base station using XBEE. Base station uses a ps2 controller to move the robot and a keypad to select gait

Outline

- Purpose
- Application
- Design Overview
- Technical Challenges

Purpose

Purpose

Goal: Create a fun game with a novel mobility scheme

Potential Use Cases:

- Exploring difficult terrain
- Research different modes of walking (such as different number of legs used)
- Games/Fun

Our Use Case:

A game where users have to navigate through a field with obstacles.

We will time users to induce competition.

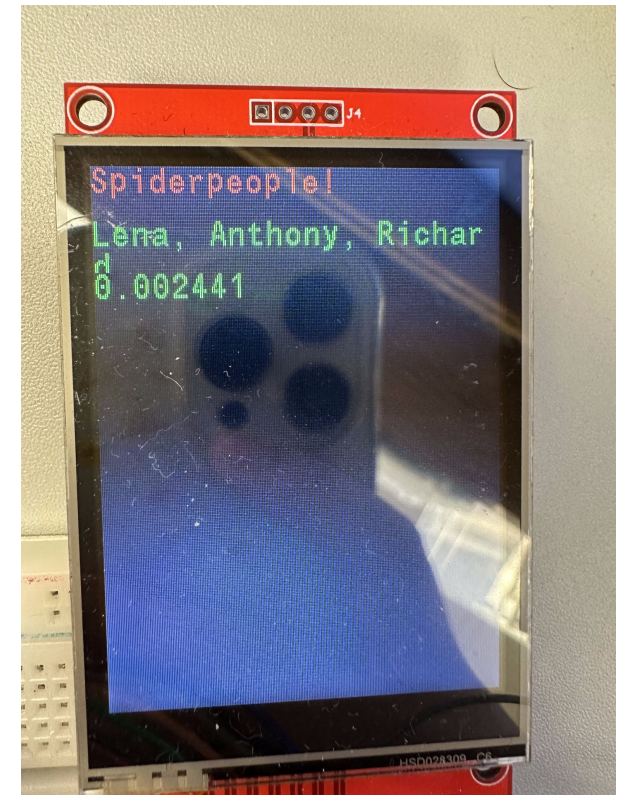
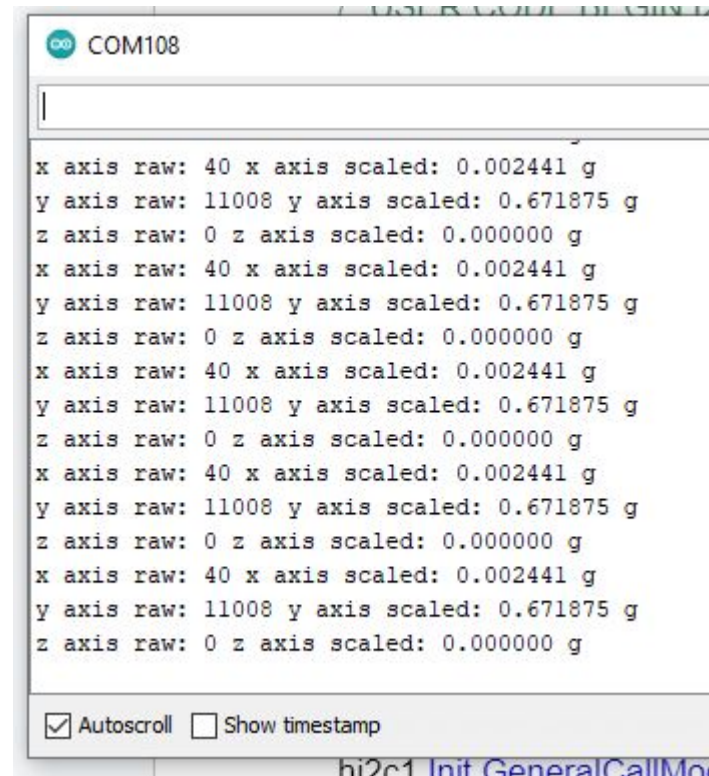


QWOP is an example of a game with a novel mobility scheme.

Application

Application from User's POV

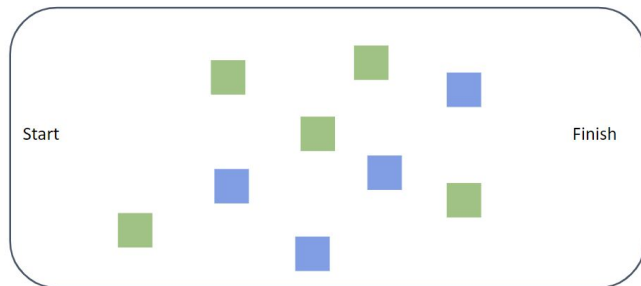
- Controller with analog sticks that control Spider's movement
- LCD display that shows the tilt of Spider and mode of walking enabled
- Keypad to switch mode of walking of the Spider



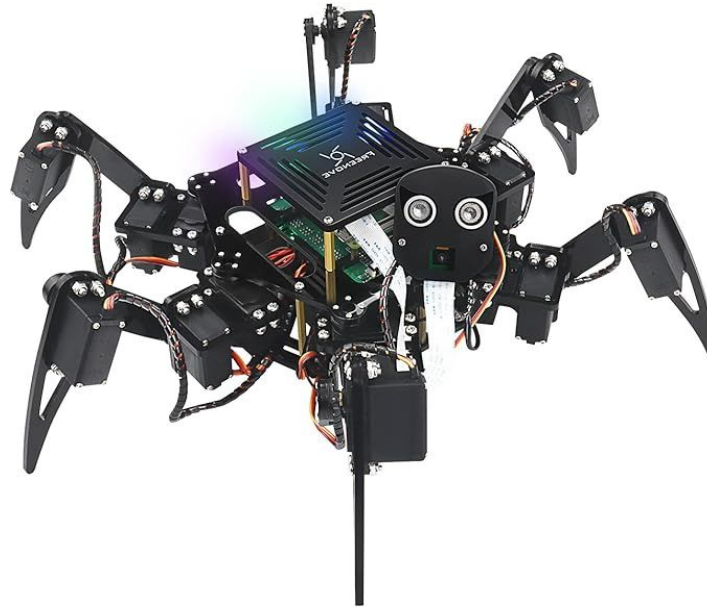
Design Overview

Major Components

Field



Spider



Base Station

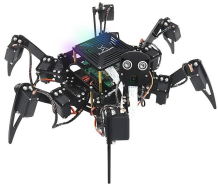


Freeove Hexapod Kit

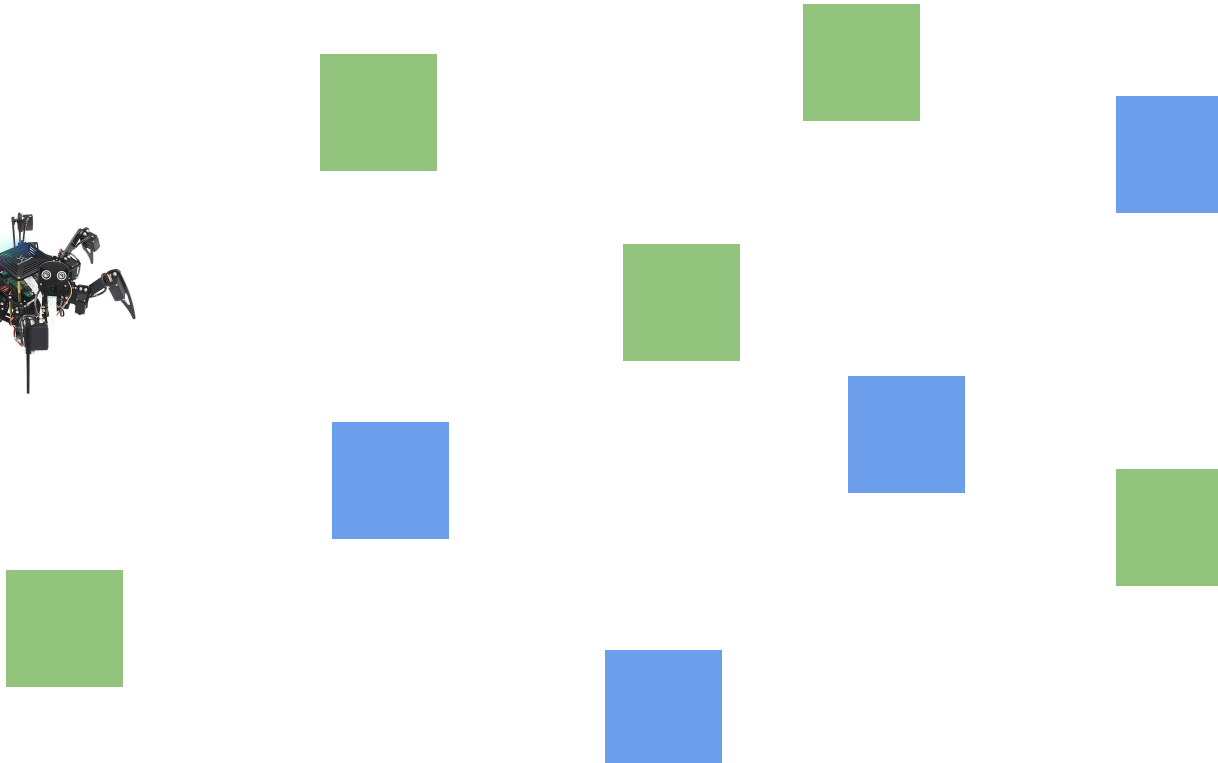
Field

Possible obstacles: Immovable blocks and unique terrain (for example: sand, gravel, or dirt), carrying objects

Start

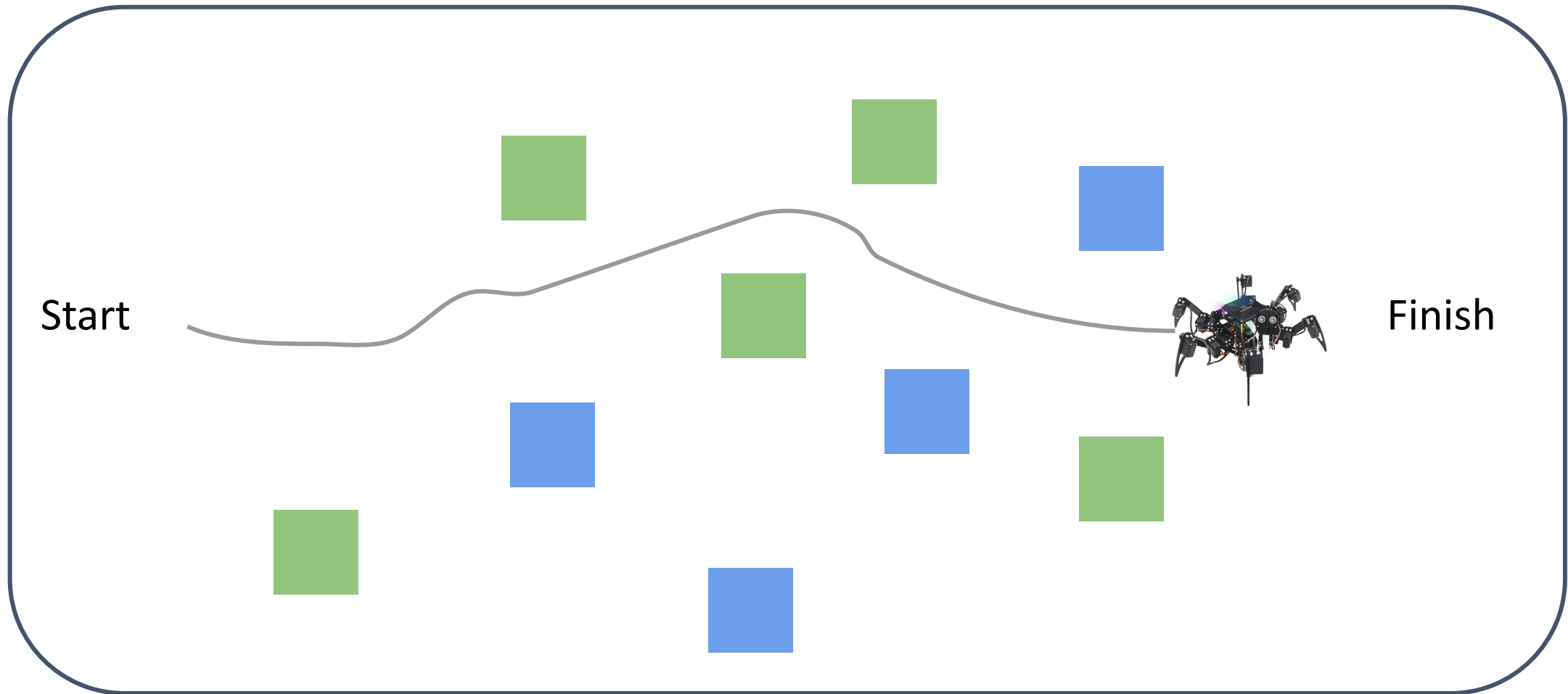


Finish



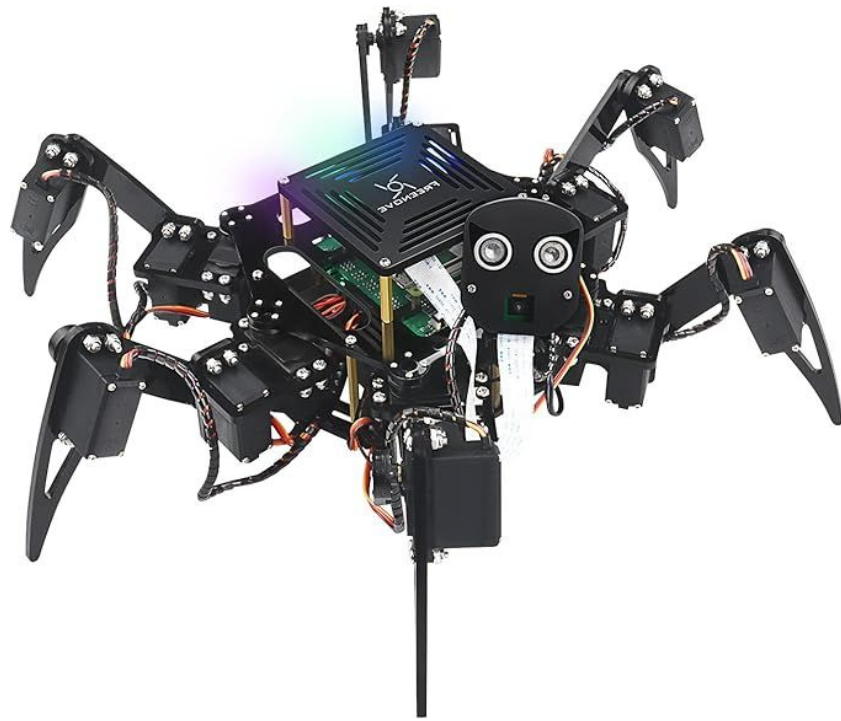
Field

Possible obstacles: Immovable blocks and unique terrain (for example: sand, gravel, or dirt), carrying objects



Spider

Amazon hexapod kit with additional components



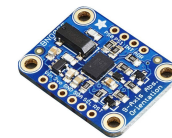
STM32 Nucleo: processor



Servos (18X) + breakout board:
leg movement

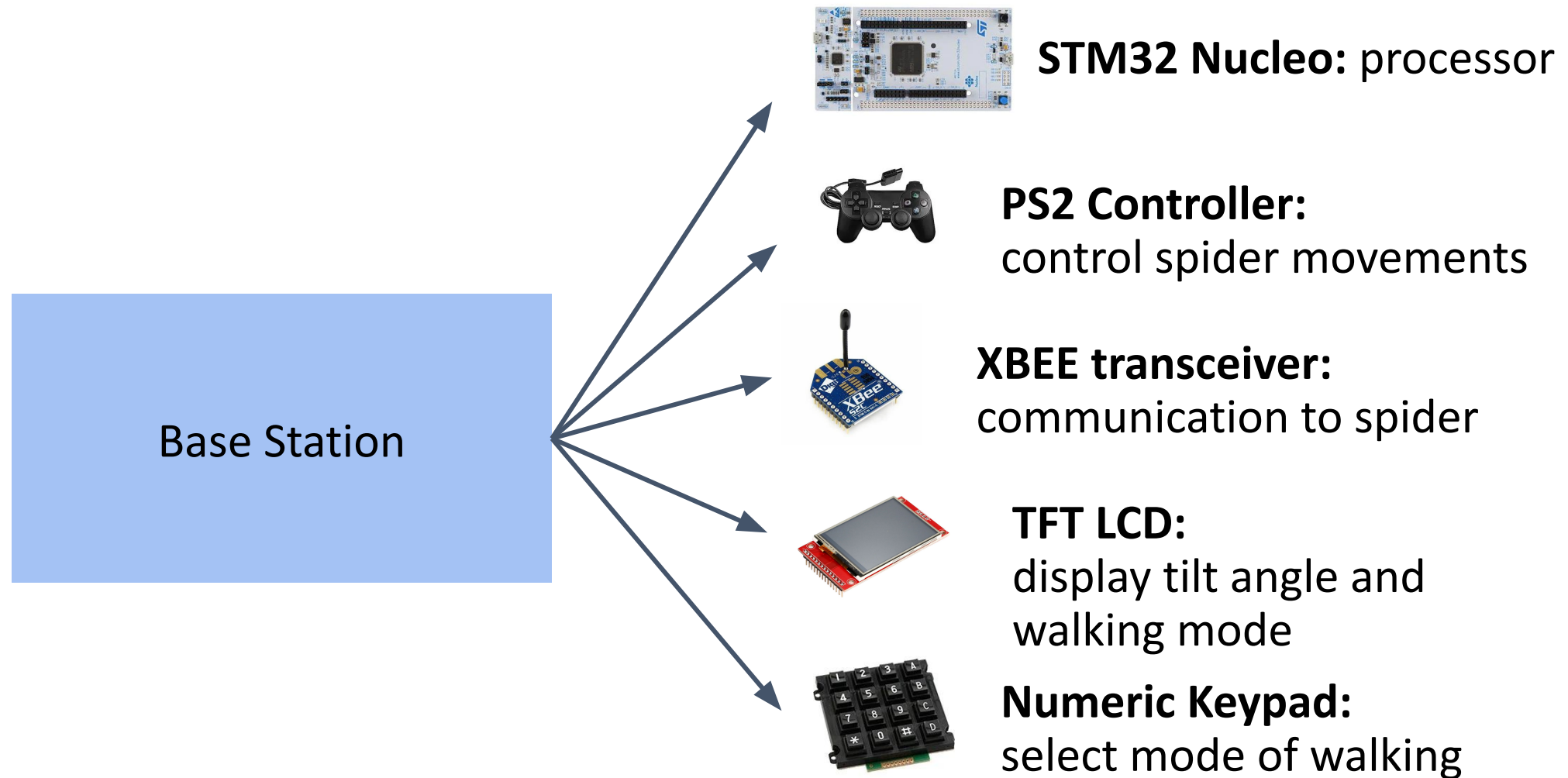


XBEE transceiver:
communication to base station

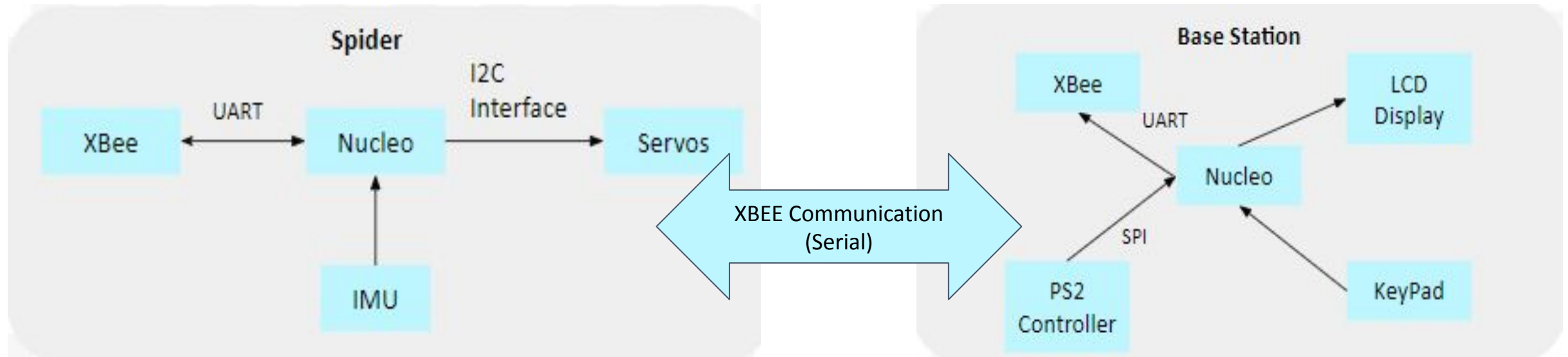


IMU: tilt angle

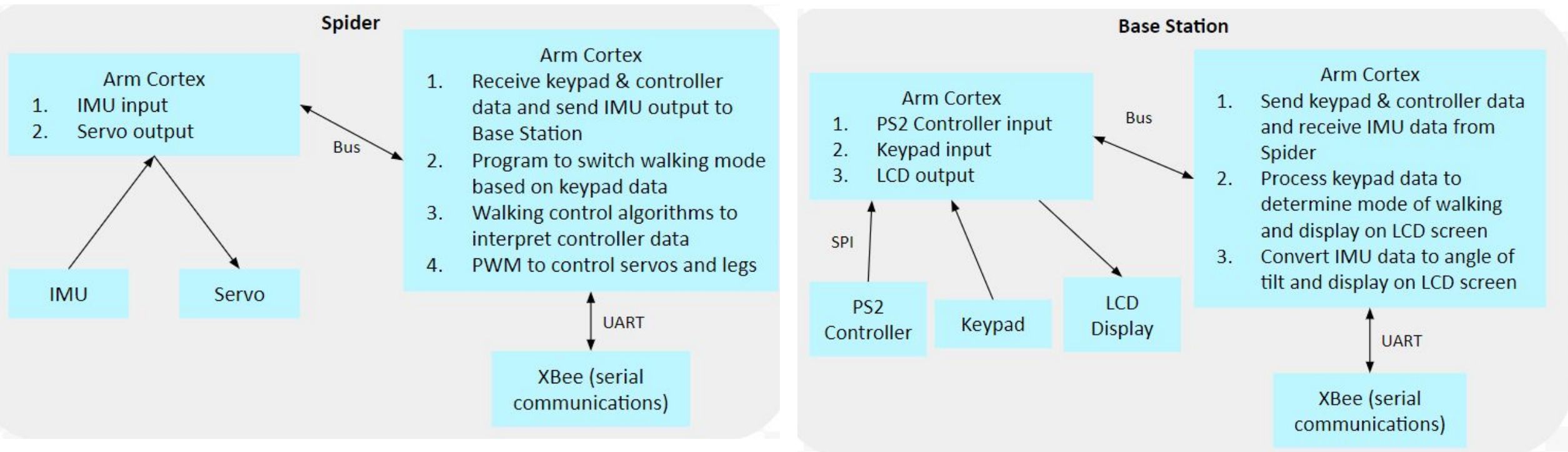
Base Station



Functional Diagram



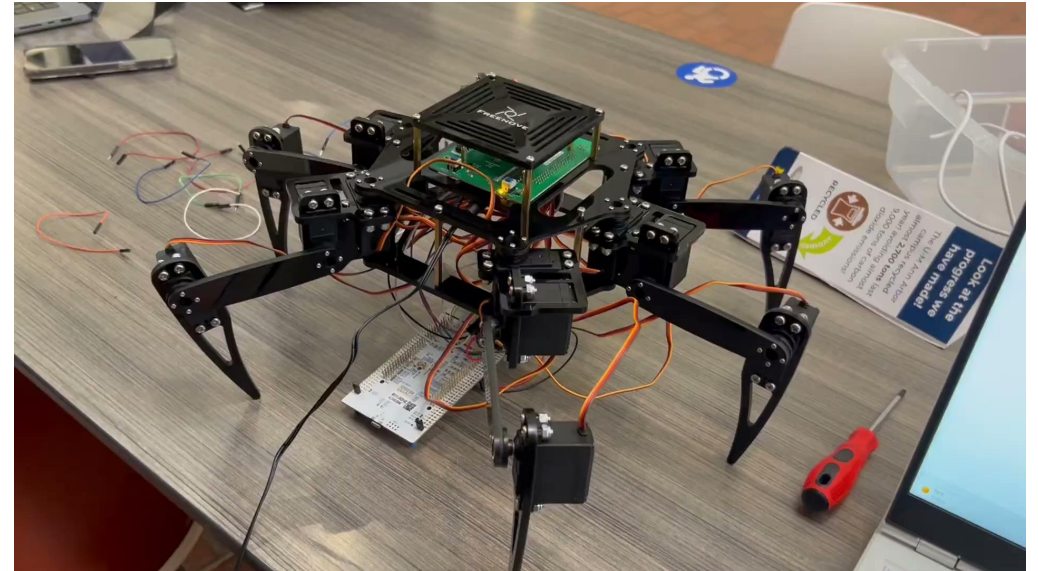
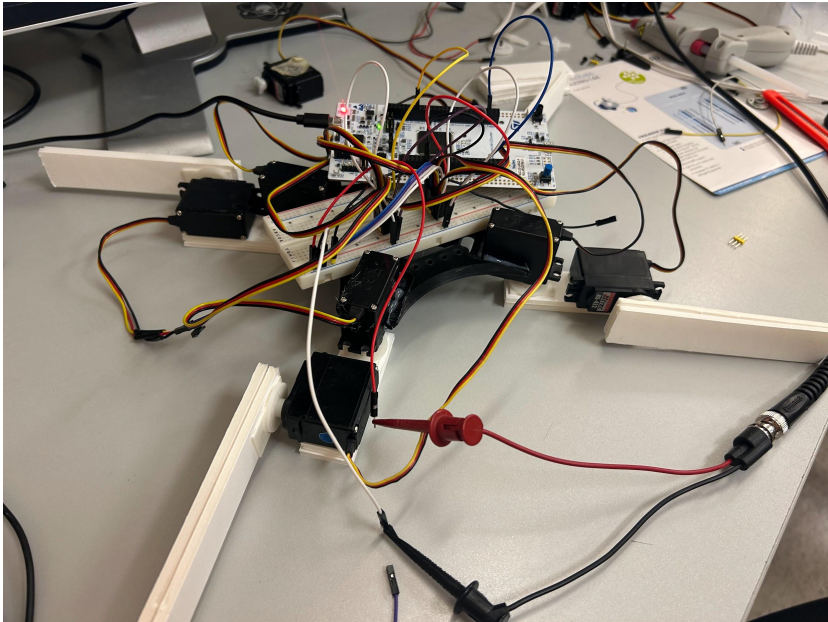
Component Diagram



Technical Challenges

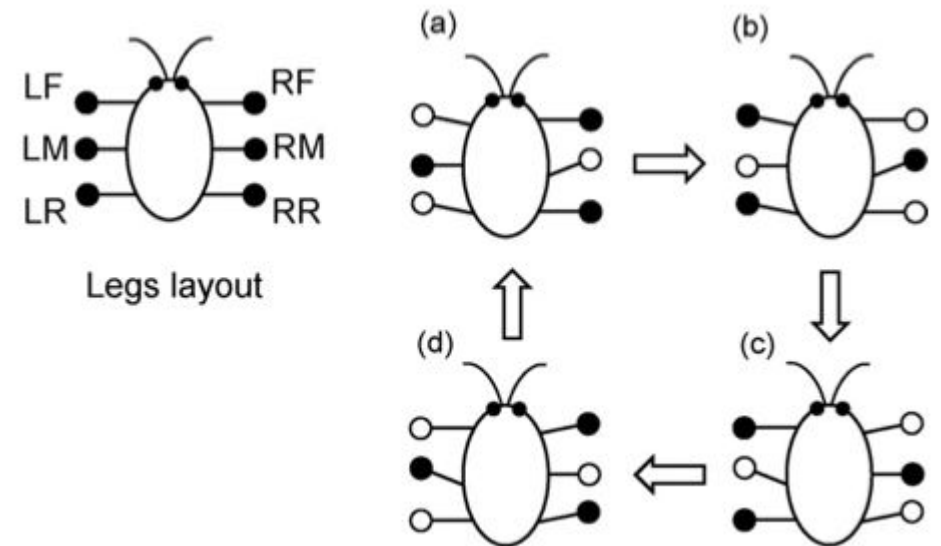
Technical Challenges

Plastic servos vs Metal servos



Technical Challenges

Emulating different walking styles



Source:

<https://tripodgaitasamodelforrobots.weebly.com/tripod-walking-gait.html>

Technical Challenges

TFT LCD Code

- Writing custom firmware
- Start up sequence
- Fonts and Colors
- Library use in parallel for integration





Thank you!
